

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The function $f(w) = 6w^2$ gives the area of a rectangle, in square feet (ft^2), if its width is w ft and its length is 6 times its width. Which of the following is the best interpretation of $f(14) = 1,176$?

Answer

- A. If the width of the rectangle is 14 ft, then the area of the rectangle is 1,176 ft^2 .
- B. If the width of the rectangle is 14 ft, then the length of the rectangle is 1,176 ft.
- C. If the width of the rectangle is 1,176 ft, then the length of the rectangle is 14 ft.
- D. If the width of the rectangle is 1,176 ft, then the area of the rectangle is 14 ft^2 .

Correct Answer: A

Rationale

Choice A is correct. The function f gives the area of the rectangle, in ft^2 , if its width is w ft. Since the value of $f(14)$ is the value of $f(w)$ if $w = 14$, it follows that $f(14) = 1,176$ means that $f(w)$ is 1,176 if $w = 14$. In the given context, this means that if the width of the rectangle is 14 ft, then the area of the rectangle is 1,176 ft^2 .

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from interpreting $f(w)$ as the width, in ft, of the rectangle if its area is $w \text{ ft}^2$, rather than as the area, in ft^2 , of the rectangle if its width is w ft.